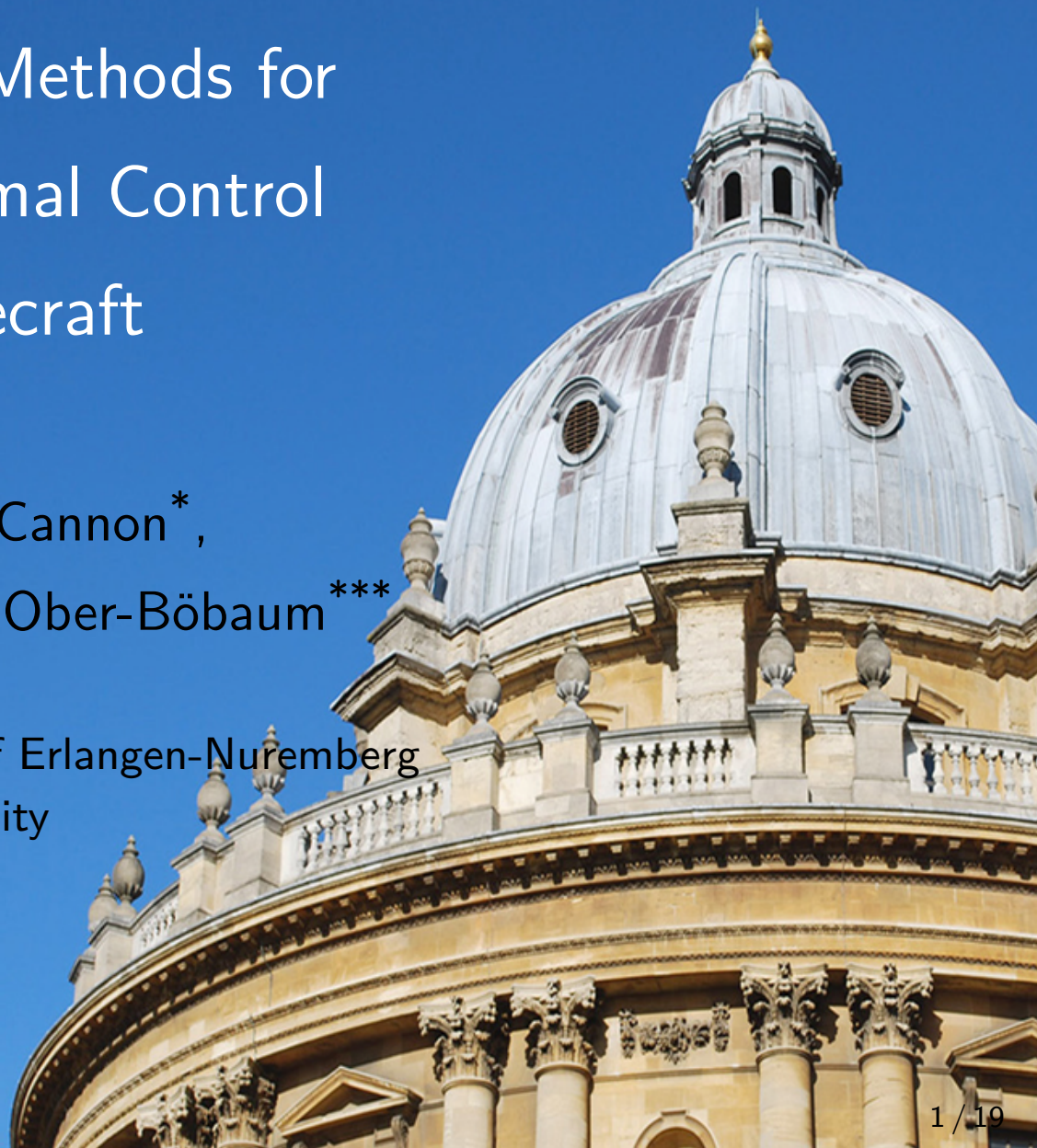
Multirate Variational Methods for Simulation and Optimal Control of Flexible Spacecraft

Yana Lishkova[*], Mark Cannon[*],
Sigrid Leyendecker[**], Sina Ober-Böbaum[***]

[*] University of Oxford, [**] University of Erlangen-Nuremberg
[***] Paderborn University

Presented by Pol Mestres



Outline

- Motivation
- Proposed approach
- Mathematical model and method implementation
- Numerical results
- Future work

Motivation

Flexible spacecraft



Figure 1: Artist visualisation of the Galileo Satellite (credit: ESA)

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Flexible spacecraft

- Stringent positioning and safety requirements
- Limited operational resources



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$\Rightarrow$ Lightweight structure designs

$\Rightarrow$ Induced vibrations degrade performance and pointing accuracy, and can even cause structural damage



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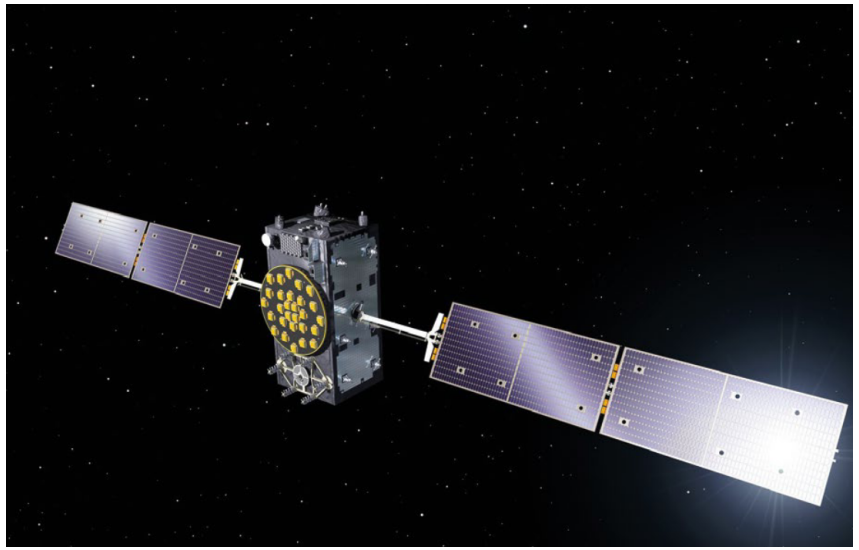


Figure 1: Artist visualisation of the Galileo Satellite (credit: ESA)

$\therefore$ Need for an efficient control method that maximizes system's performance while respecting all hard safety-critical constraints

Proposed Approach

Multirate Discrete Mechanics and Optimal Control (Multirate DMOC)



Figure 2: Artist visualisation of the Galileo Satellite (credit: ESA)

[11] Gail, S. Ober-Blöbaum, and S Leyendecker, 'Variational multirate integration in discrete mechanics and optimal control,' in *Proceedings of ECCOMAS*, pages 1–4, 2017

Proposed Approach

Multirate Discrete Mechanics and Optimal Control (Multirate DMOC)

- Exploiting **hidden structure** (conservation laws) encoded in the equations of motion, using a *variational formulation*
- Exploiting the **time scales** of the dynamics, via *multirate discretization*



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$\therefore$ to provide **high-fidelity simulation at a reduced computational cost**

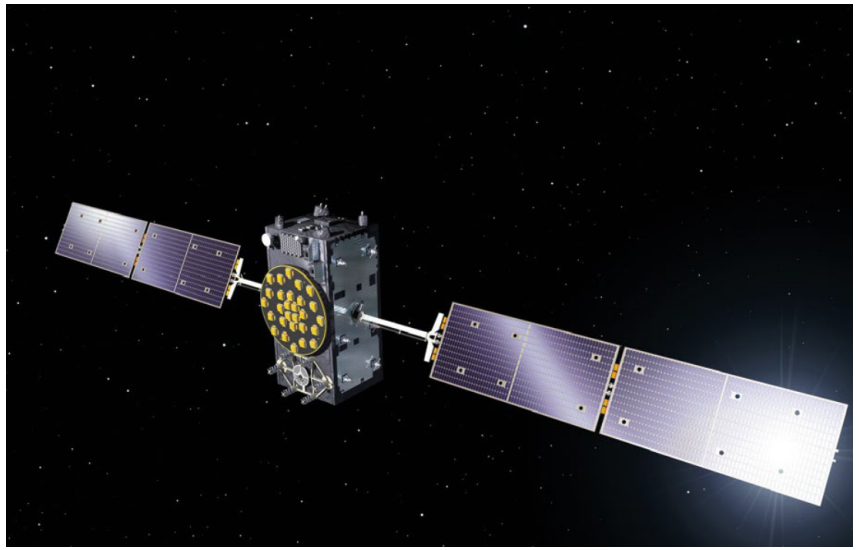


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A Typical Optimal Control Problem

A typical problem takes the form:

$$\min_{u(\cdot)} \int_0^T C\Big(x(t), u(t)\Big)\, dt$$

subject to

$$x(0) = x_{\text{init}}$$

$$\dot{x}(t) = f\Big(x(t), u(t)\Big)$$

$$x(t) \in \mathcal{X}, \quad u(t) \in \mathcal{U}$$

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$\Rightarrow$ As a continuous problem, this needs infinitely many constraints on x, u for every $t \in [0, T]$ and is thus not tractable on a computer. One way to solve it: discretise state & control on a finite time grid.

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A discretised, finitely-constrained version:

$$\min_{\{u_k\}} \sum_{k=0}^{N-1} C\big(x_k, u_k\big)$$

subject to

$$x_0 = x_{\text{init}}$$

$$x_{k+1} = f(x_k, u_k)$$

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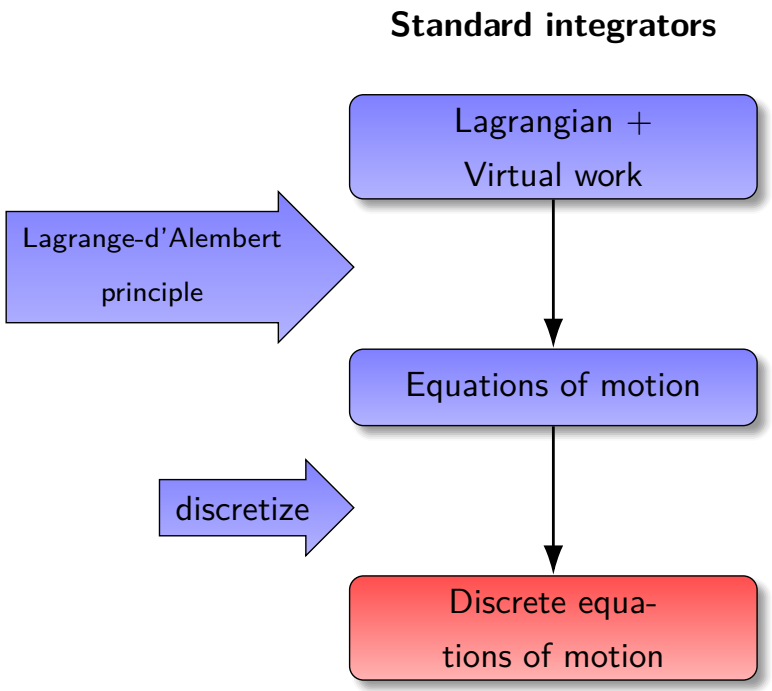
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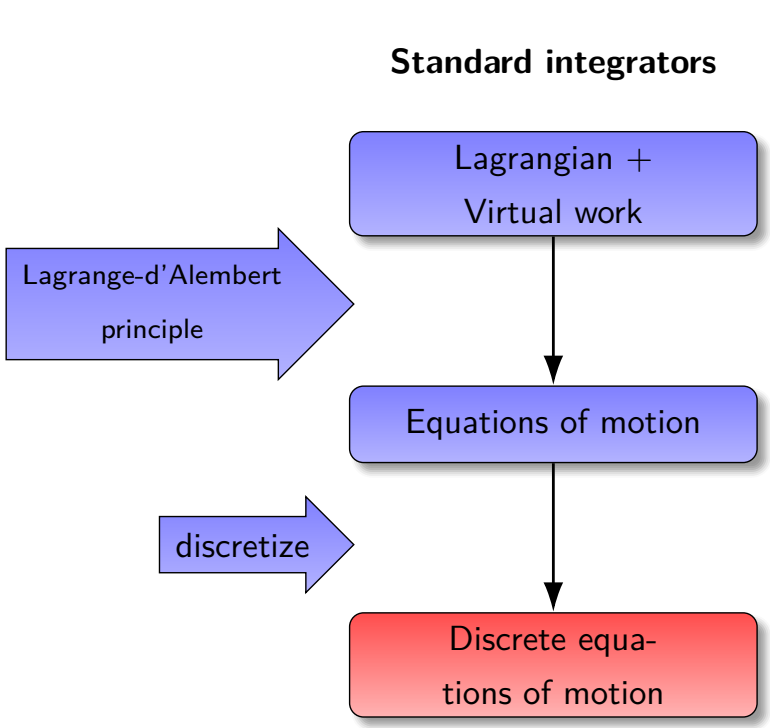
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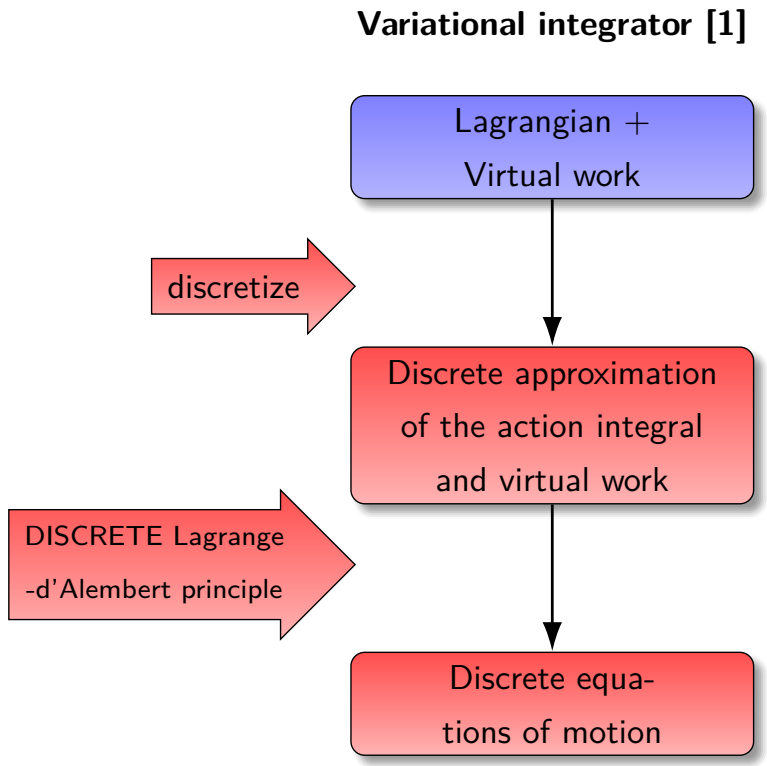
$\therefore$ Hidden structure (conservation laws) lie in the continuous dynamics $\dot{x} = f(x, u)$ and can be preserved by discretising in a special (variational) way, yielding **the DMOC** approach.

Variational integrators



Variational integrators





[1] J. E. Marsden and M. West, "Discrete mechanics and variational integrators," Acta Numerica, Vol. 10, No. 1, 2001, pp. 357–514.

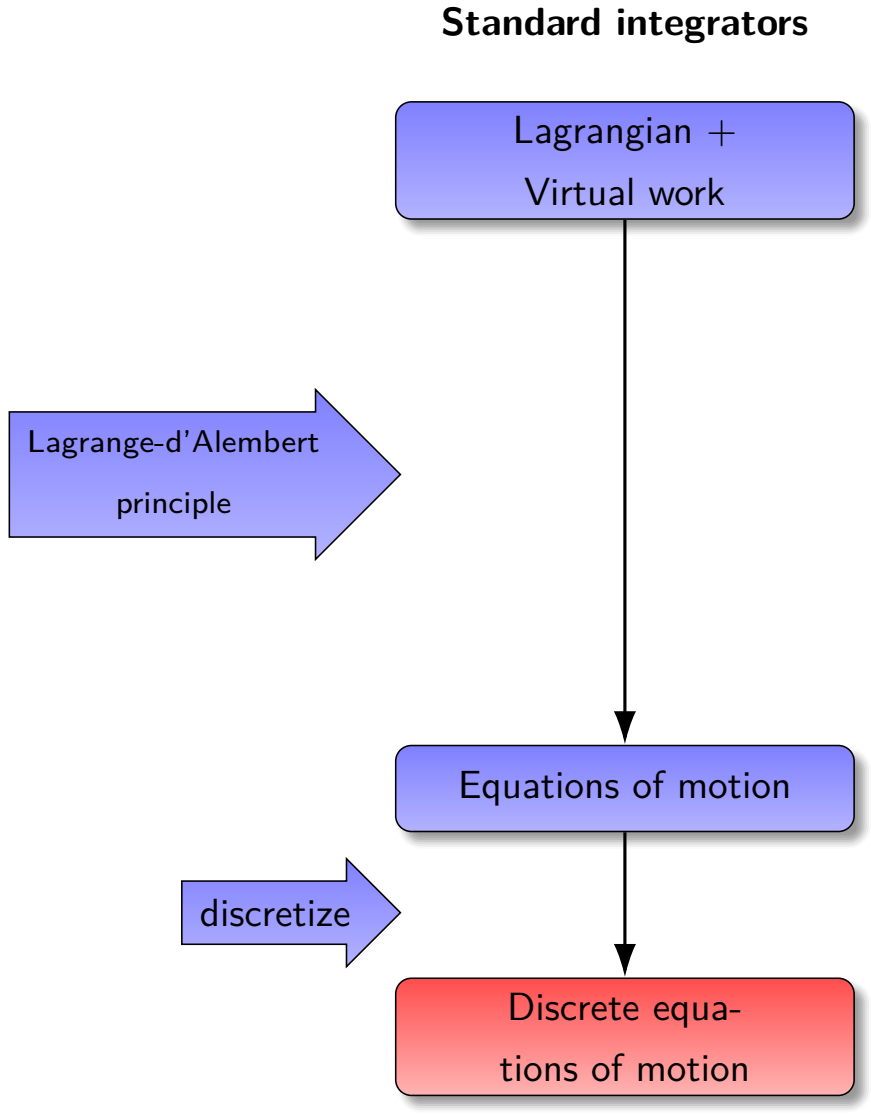
Standard integrators

Action integral

$$\int_0^T L(q(t), \dot{q}(t))dt$$

Virtual work

$$\int_0^T f(q(t), \dot{q}(t)) \cdot \delta q(t)dt$$



Standard integrators

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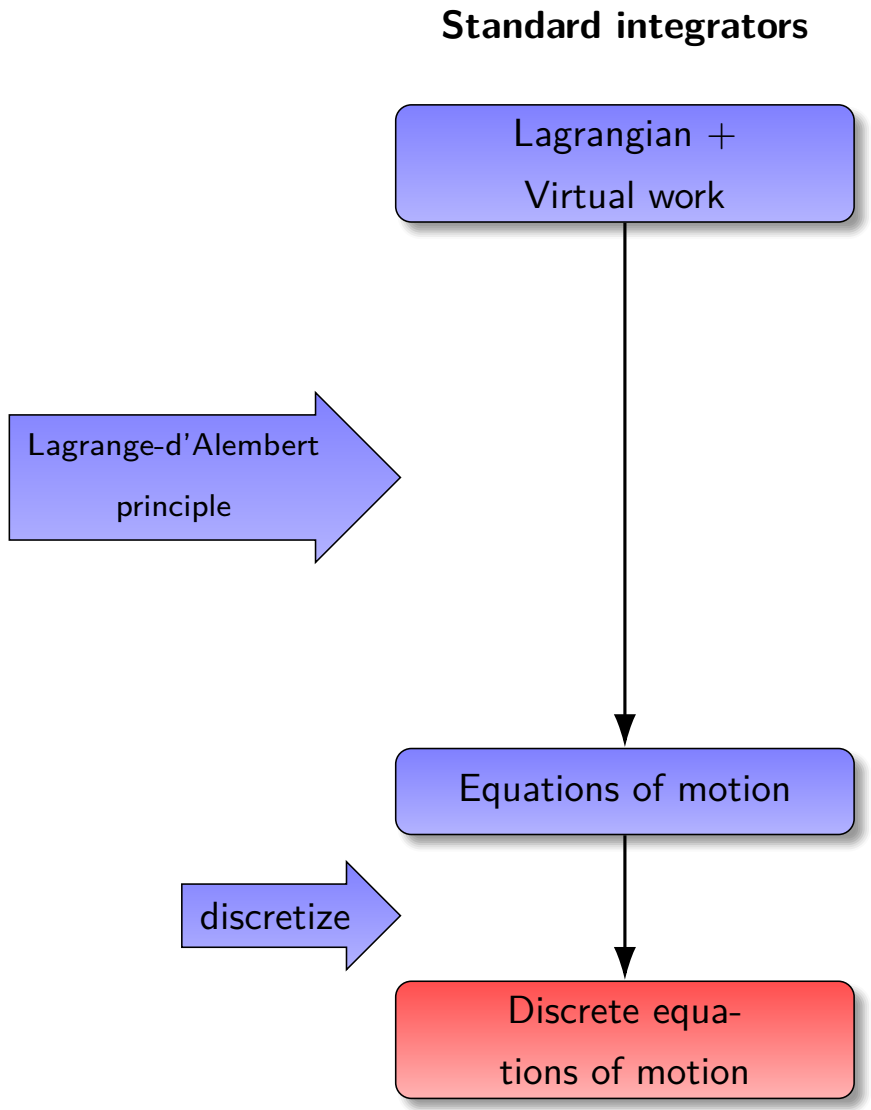
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Lagrange-d'Alembert principle for all variations δq with $\delta q(0) = \delta q(T) = 0$:

$$\delta \int_0^T L(q(t), \dot{q}(t)dt + \int_0^T f(q(t), \dot{q}(t)) \cdot \delta q(t)dt = 0$$



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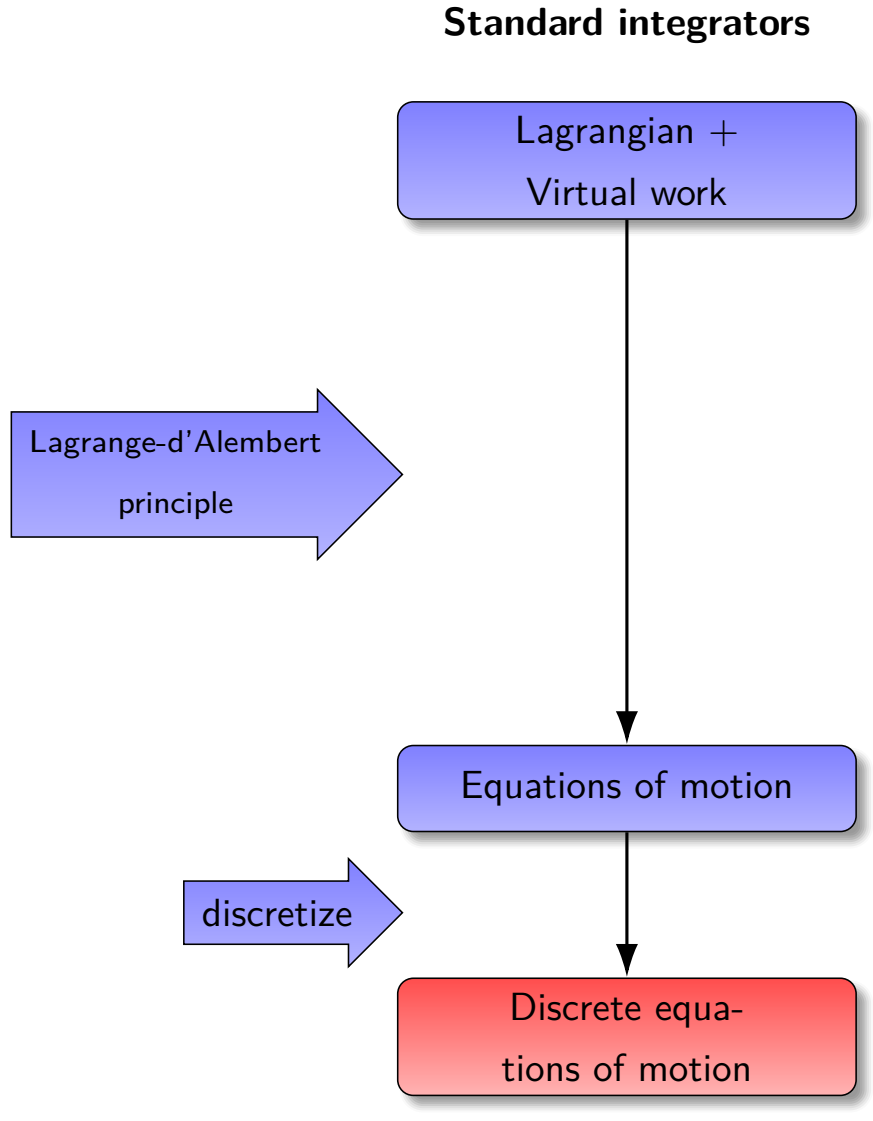
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leads to the the forced Euler-Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = f(q, \dot{q})$$

These can be discretized.



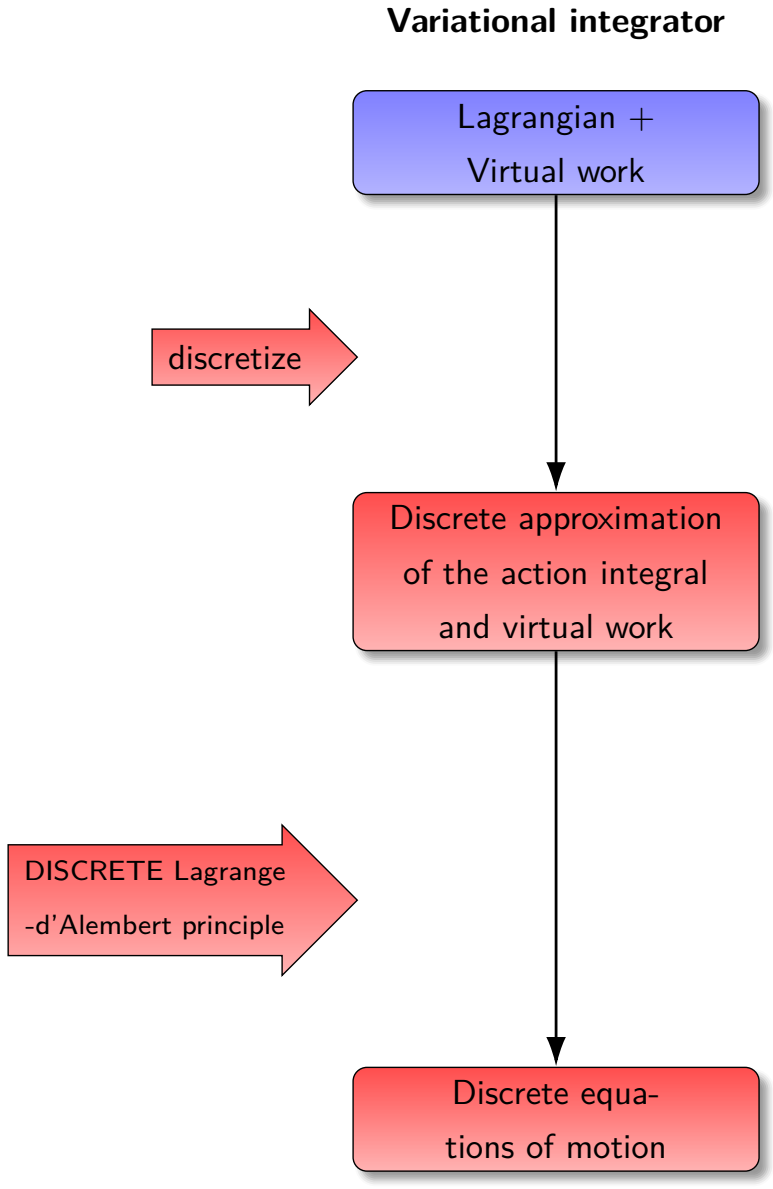
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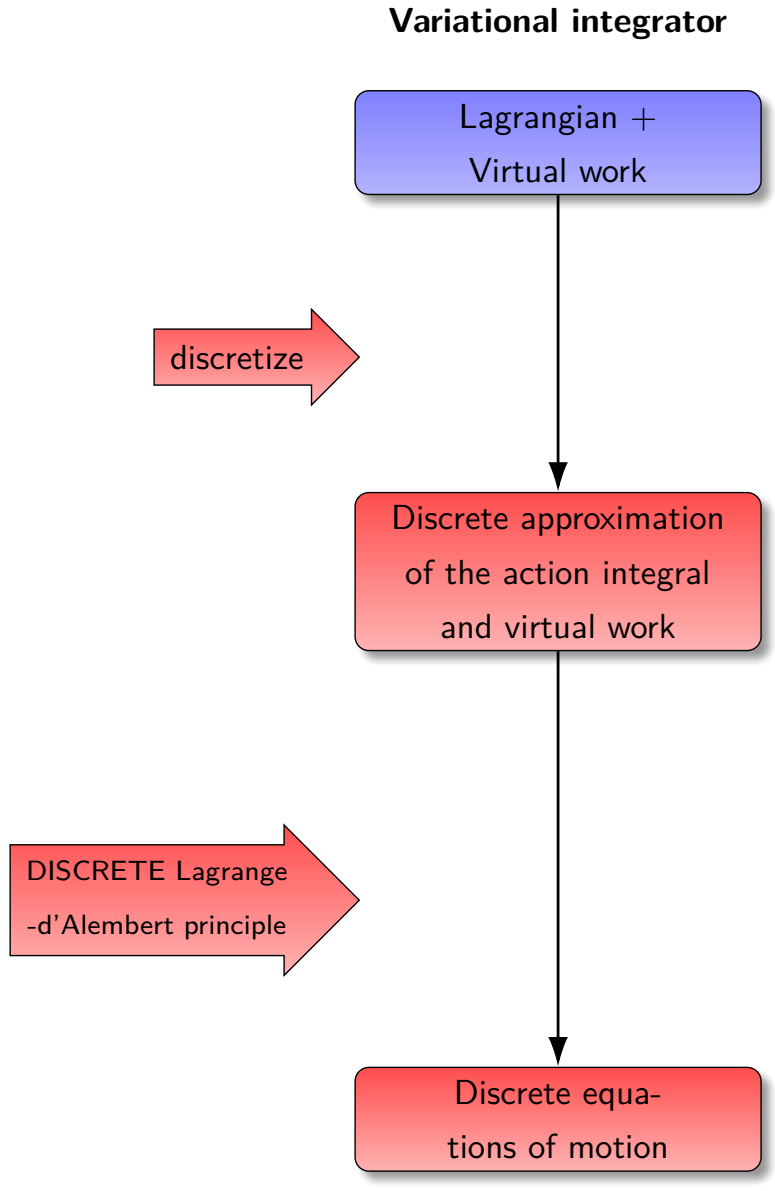
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The *discrete* Lagrangian

$$L_d(q_k, q_{k+1}) \approx \int_{k\Delta t}^{(k+1)\Delta t} L(q(t), \dot{q}(t))dt$$

Virtual work approximation by the left and right *discrete* forces

$$f_k^- \cdot \delta q_k + f_k^+ \cdot \delta q_{k+1} \approx \int_{k\Delta t}^{(k+1)\Delta t} f(t) \cdot \delta q(t)dt$$



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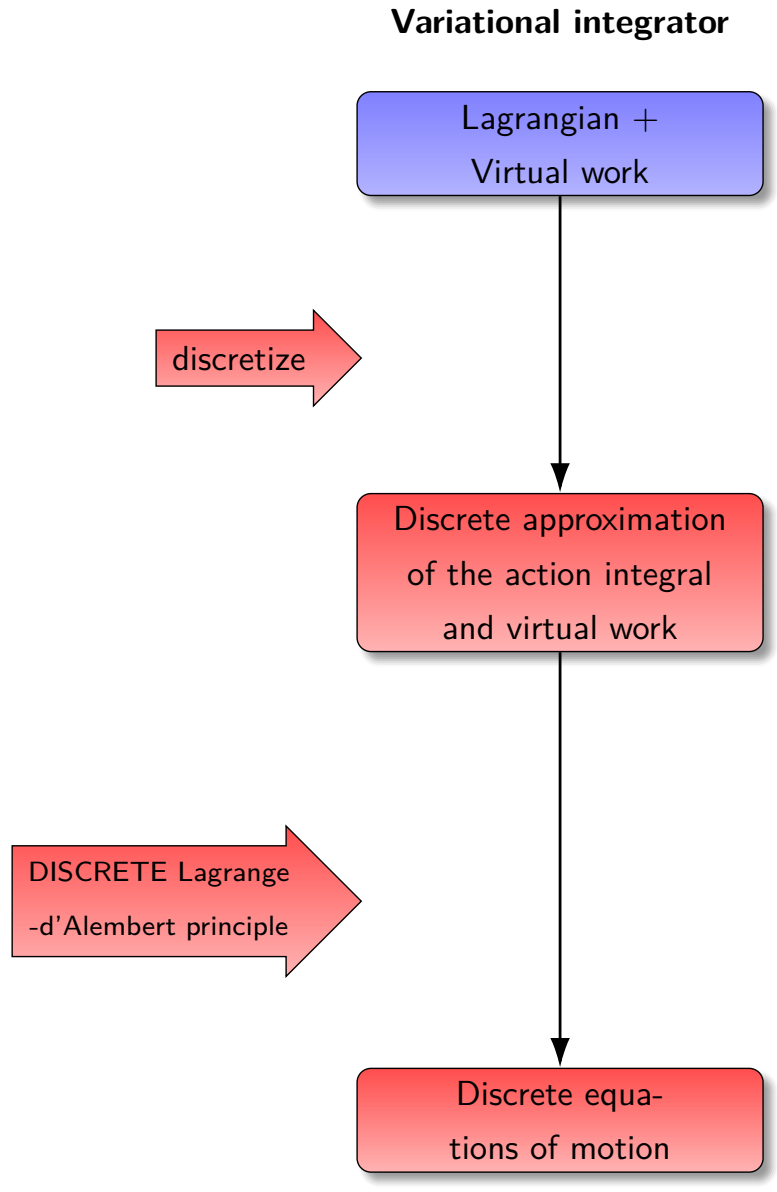
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Applying the *discrete* Lagrange-d'Alembert principle for all variations $\{q_k\}_{k=0}^N$ with $\delta q_0 = \delta q_N = 0$ $(N = T/\Delta t)$

$$\delta \sum_{k=0}^{N-1} L_d(q_k, q_{k+1}) + \sum_{k=0}^{N-1} (f_k^- \cdot \delta q_k + f_k^+ \cdot \delta q_{k+1}) = 0$$



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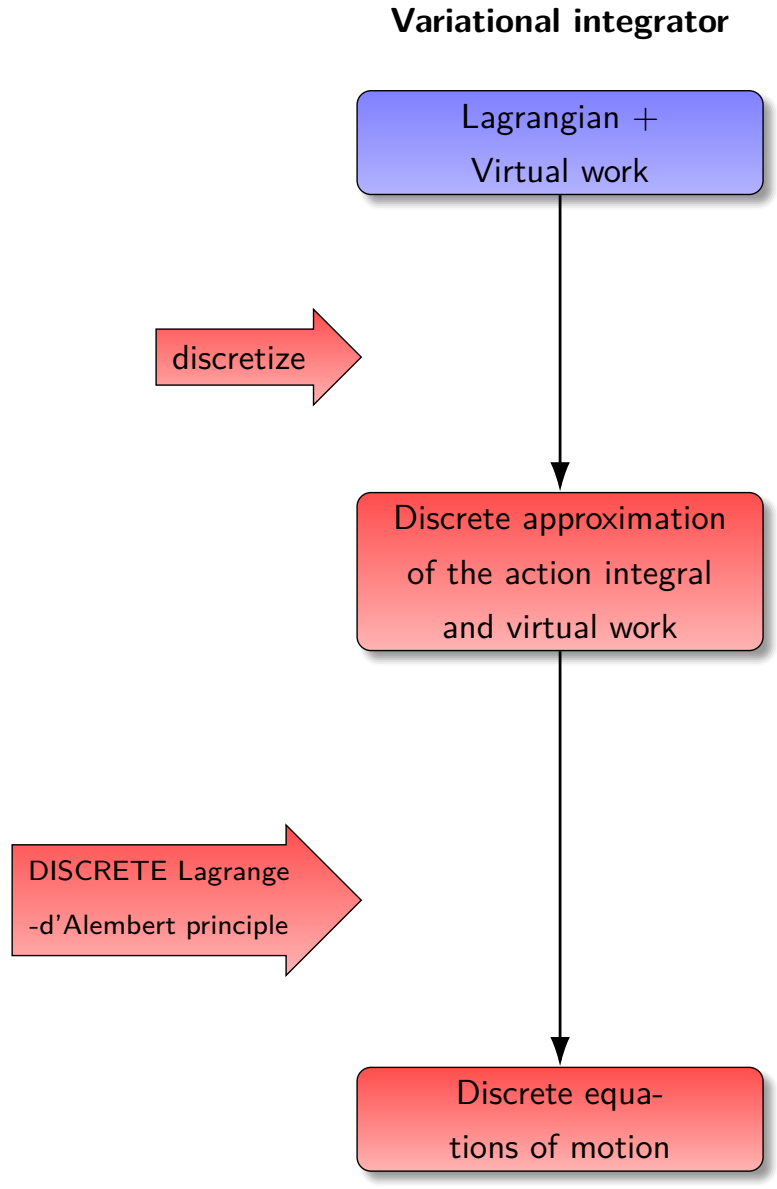
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to obtain the *discrete* forced Euler-Lagrange equations

$$D_2 L_d(q_{k-1}, q_k) + D_1 L_d(q_k, q_{k+1}) + f_{k-1}^+ + f_k^- = 0$$



Two Equivalent Forms: LF and HF

Lagrangian formulation (LF) — configuration-only:

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Example quadrature (midpoint rule):

$$L_d(q_k, q_{k+1}) = L\left(\tfrac{q_{k+1}+q_k}{2}, \tfrac{q_{k+1}-q_k}{\Delta t}\right), \qquad f_k^{\pm} = f\left(\tfrac{q_{k+1}+q_k}{2}, \tfrac{q_{k+1}-q_k}{\Delta t}\right)$$

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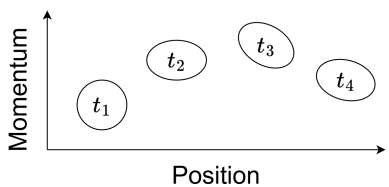
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Structure-preserving:

- momentum preservation
- symplecticity preservation
- energy preservation up to an oscillatory bounded error



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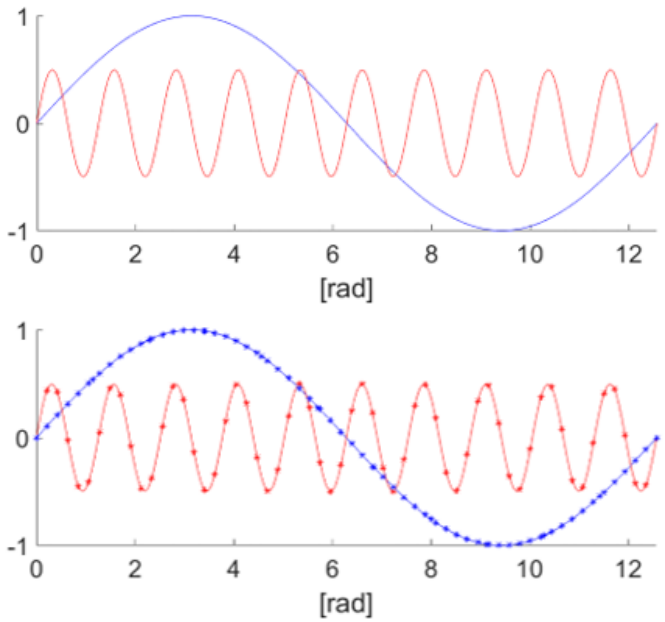
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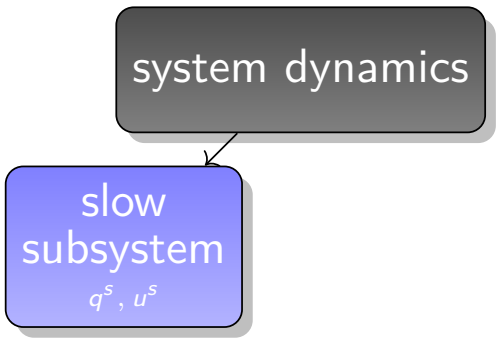
Next: exploit the timescales of the system dynamics to obtain computational savings for comparable accuracy.

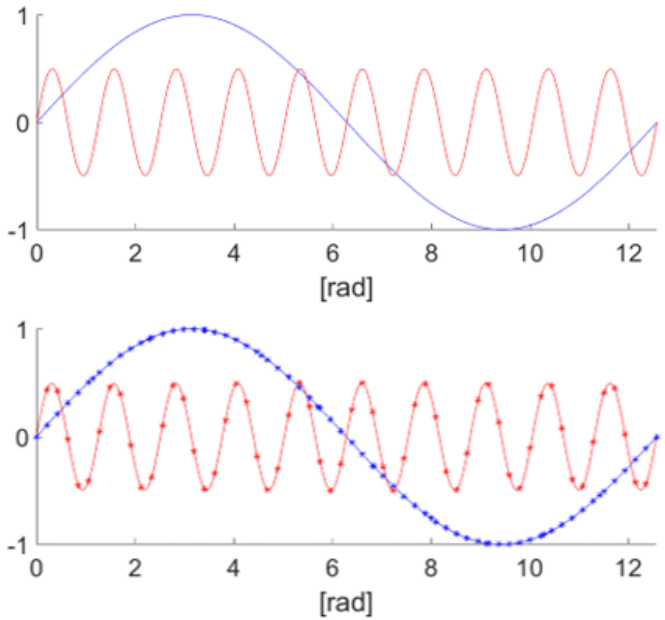
Multirate Variational Integrators: Splitting Slow & Fast

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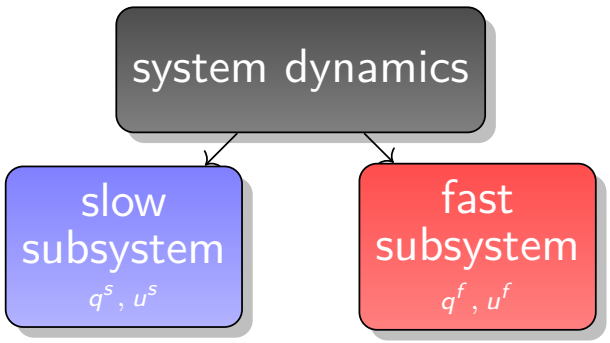


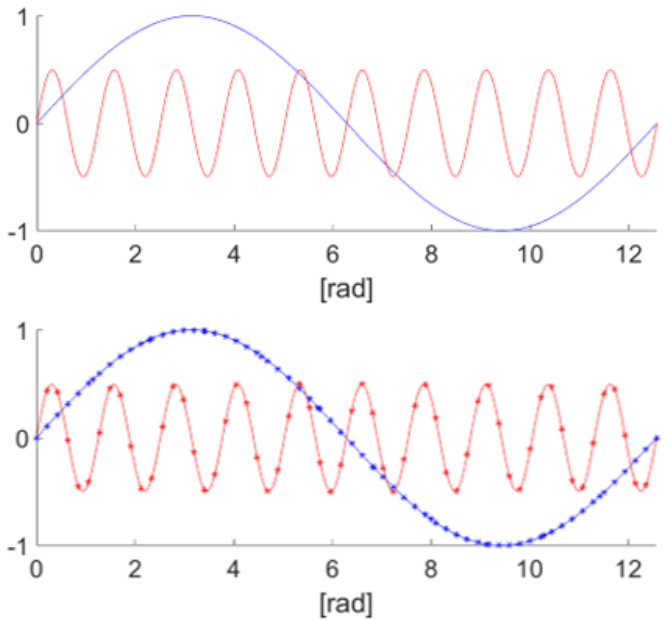
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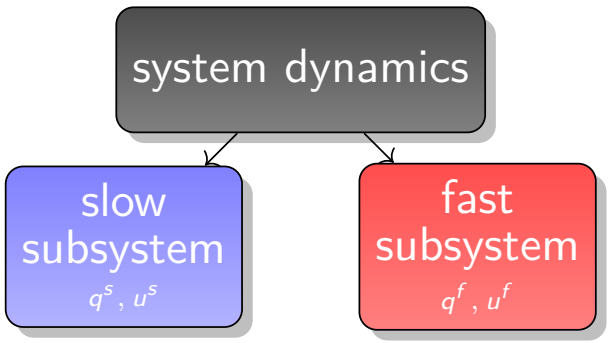


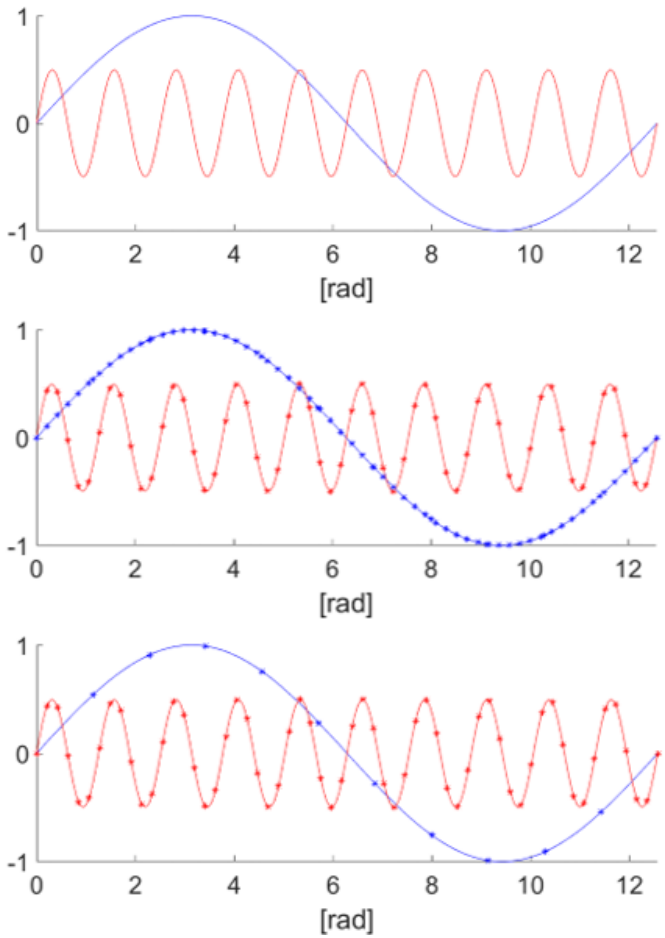
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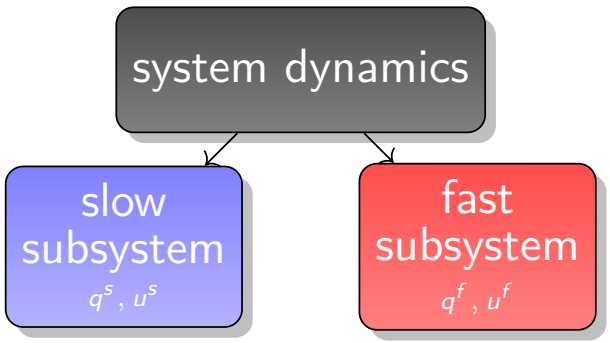


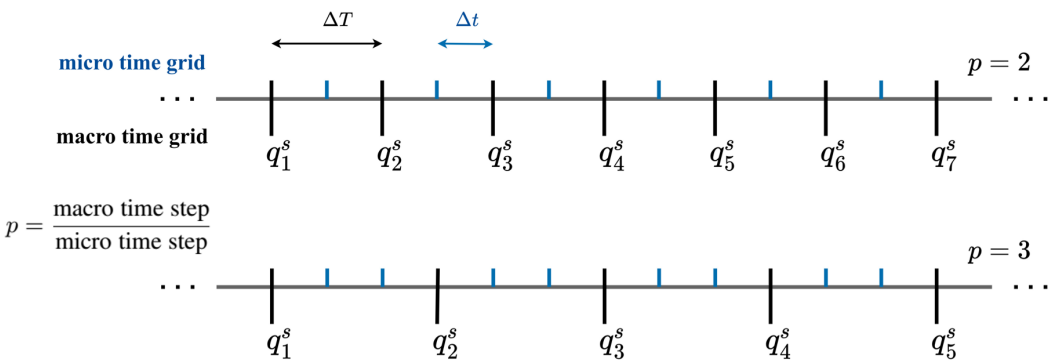
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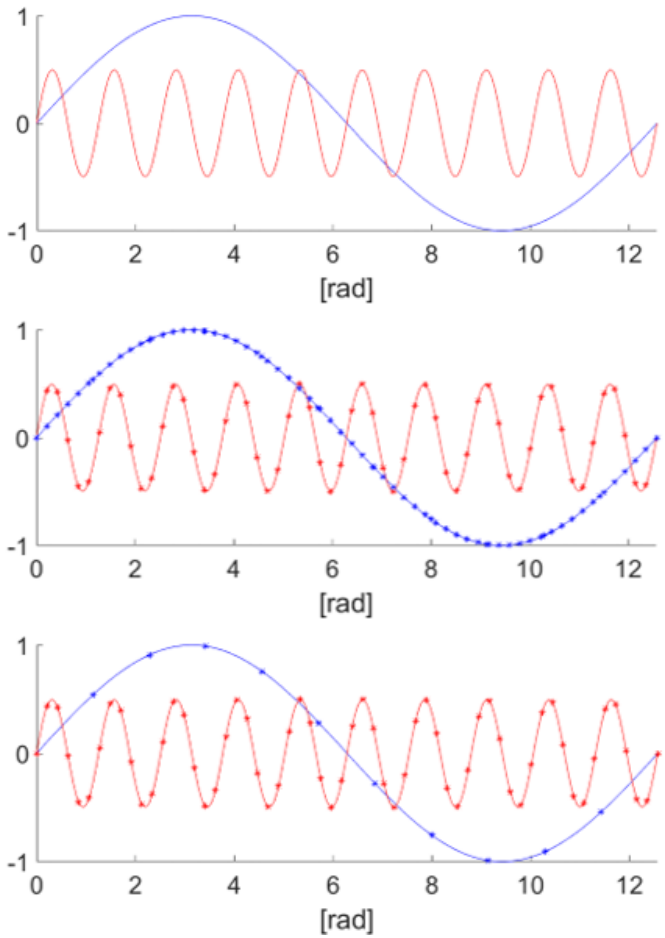




Multirate Variational Integrators: Splitting Slow & Fast







[2] T. Gail, S. Ober-Blöbaum, and S. Leyendecker, "Variational multirate integration in discrete mechanics and optimal control,"

Encoding the Dynamics: Multirate DMOC

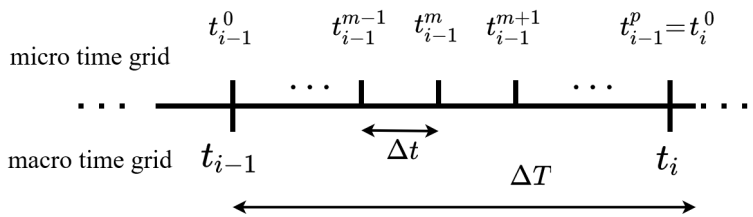
$$\min_{\mathbf{u},\,\mathbf{x}} \sum_{i=0}^{N} \left(\|x_i - x_k^r\|_Q^2 + \|u_i - u_i^r\|_R^2 \right)$$

subject to, for $i = 1, \ldots, N-1$

$$x_0 = x_{\text{init}}, \qquad x_{i+1} = f(x_i, u_i), \quad x_i \in \mathcal{X}, \quad u_i \in \mathcal{U}$$

with:

$$x_i = \begin{bmatrix} q_i^s \\ q_i^{f,0} \\ \vdots \\ q_i^{f,p} \end{bmatrix}, \; u_i = \begin{bmatrix} u_i^s \\ u_i^{f,0} \\ \vdots \\ u_i^{f,p} \end{bmatrix}$$



Encoding the Dynamics: Multirate DMOC

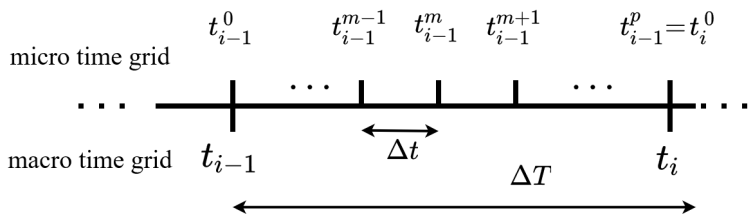
$$\min_{\mathbf{u},\,\mathbf{x}} \sum_{i=0}^{N} \left(\|x_i - x_k^r\|_Q^2 + \|u_i - u_i^r\|_R^2 \right)$$

subject to, for $i = 1, \ldots, N-1$

$$x_0 = x_{\text{init}}, \qquad x_{i+1} = f(x_i, u_i), \quad x_i \in \mathcal{X}, \quad u_i \in \mathcal{U}$$

with:

$$x_i = \begin{bmatrix} q_i^s \\ q_i^{f,0} \\ \vdots \\ q_i^{f,p} \end{bmatrix}, \; u_i = \begin{bmatrix} u_i^s \\ u_i^{f,0} \\ \vdots \\ u_i^{f,p} \end{bmatrix}$$



Multirate Discrete Mechanics and Optimal Control (Multirate DMOC) — in LF or HF form.

[11] Gail, S. Ober-Blöbaum, and S Leyendecker, 'Variational multirate integration in discrete mechanics and optimal control,' in *Proceedings of ECCOMAS*, pages 1–4, 2017

Multirate variational optimal control

Multirate DMOC:

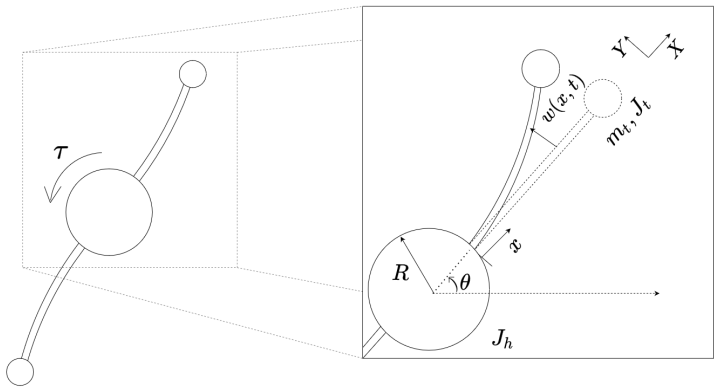
- ✓ Structure-preserving → *inherits the conservation properties of the continuous-time model*
- ✓ No velocity simulation needed → *fewer optimisation variables*
- ✓ No decoupling of the slow/fast equations of motion → *applies straightforwardly to linear and nonlinear models*
- ✓ Uniform control strategy for fast & slow dynamics → *potential for improved optimality and constraint handling*
- ✓ Reduces the number of equality constraints and unknowns → *significant computational savings*

Disadvantage: restricted to systems with a known Lagrangian representation.

[2] T. Gail, S. Ober-Blöbaum, and S Leyendecker, 'Variational multirate integration in discrete mechanics and optimal control,' in *Proceedings of ECCOMAS*, pages 1–4, 2017

Flexible satellite control

- **General linear model neglecting dissipation effects**
  - single-axis rest-to-rest rotational maneuver
  - control torque applied at the hub
- **Spatial discretization - The Assumed Modes Method [11]**



$$w(x,t) = \sum_{j=1}^{N} \phi_j(x) n_j(t), \quad \xi \in [0, L]$$

ϕ_j - assumed spatial mode shapes

η_j - time varying modal amplitudes

n - number of modes retained in the approximation

L - length of the beam

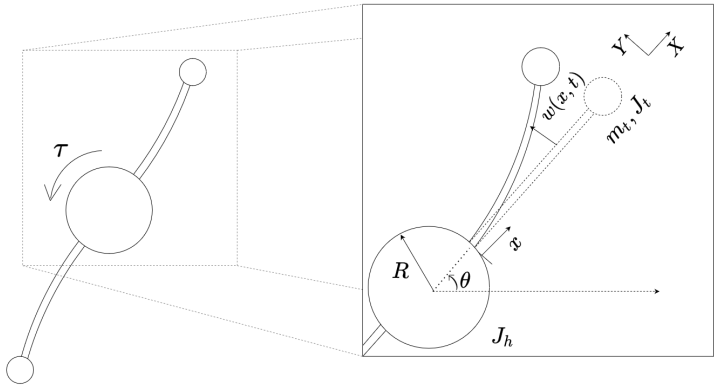
[9] **Y. Lishkova**, S. Ober-Blöbaum, M. Cannon, and S. Leyendecker, A multirate variational approach to simulation and optimal control for flexible spacecraft, Advances in the Astronautical Sciences, vol. 175, 2020.

[10] **Y. Lishkova**, S. Ober-Blöbaum, and S. Leyendecker, 'Multirate discrete mechanics and optimal control for a flexible satellite model,' in GAMM Rundbrief 2023.

[11] J. L. Junkins and Y. Kim, *Introduction to dynamics and control of flexible structures*, American Institute of Aeronautics and Astronautics, 1993.

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- **System description**

$$\xi = \begin{bmatrix} \theta \\ \eta \end{bmatrix},$$

$$\eta = [\eta_1, \eta_2, \ldots, \eta_n]^\intercal \in \mathbb{R}^{N \times 1}$$

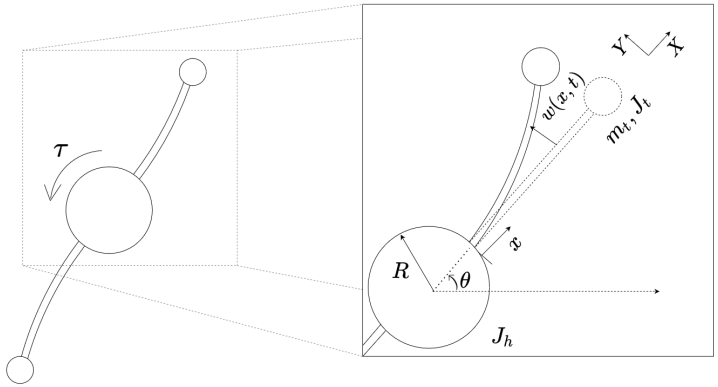
applying the Lagrange-d'Alembert Principle with

$$\mathcal{L}(\xi, \dot{\xi}) = \frac{1}{2}\dot{\xi}^\intercal M \dot{\xi} - \frac{1}{2}\xi^\intercal K \xi, \quad \delta W = f \cdot \delta\xi = \tau\delta\theta$$

$$M\ddot{\xi} + K\xi = D\tau$$

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change to modal coordinates

$$\xi = Eq, \quad E^\mathsf{T} M E = I,$$

$$E^\mathsf{T} K E = \Lambda = \mathrm{diag}[w_1^2, w_2^2, \ldots, w_{n+1}^2]$$

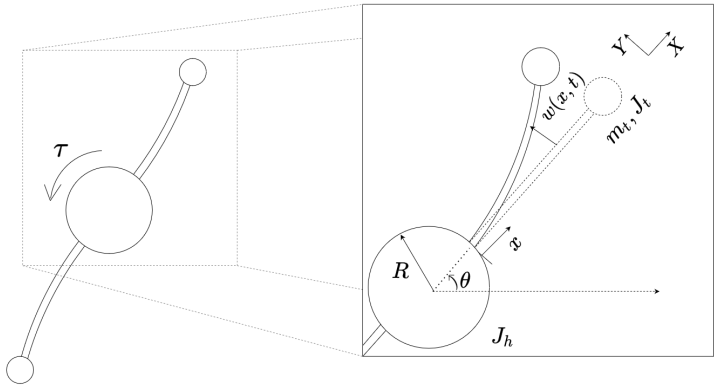
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$$M\ddot{\xi} + K\xi = D\tau$$

$$\mathfrak{L}(q, q) = \frac{1}{2}\dot{q}^\mathsf{T}\dot{q} - \frac{1}{2}q^\mathsf{T}\Lambda q, \quad f = E^\mathsf{T} D\tau$$

$$\ddot{q} + \Lambda q = E^\mathsf{T} D\tau$$

Flexible Satellite: Manoeuvre & Tracking Results

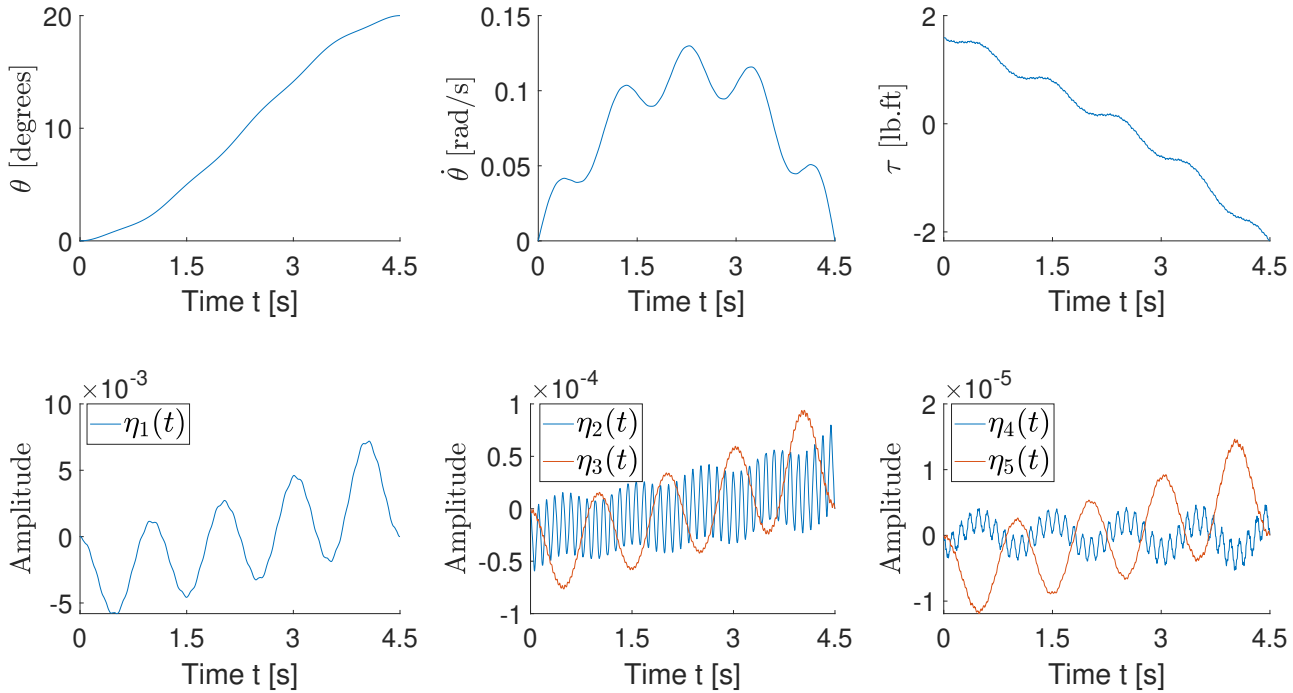


Figure 3: OCP solution with Multirate DMOC for $\theta_{t_f} = 20^\circ$, $t_f = 4.5s$, $\Delta t = 10^{-3}$ and $p = 5$

Table 1: Structural parameters used for the simulations

Hub radius	R	1.0	ft
Hub rotary inertia	J_h	8.0	slug-ft^2
Tip mass	m_t	0.156941	slug
Tip mass rotary inertia	J_t	0.0018	slug-ft^2
Beam length	L	4.0	ft
Beam height	h	6.0	in.
Beam thickness	t	0.125	in.
Beam linear density	ρA	0.0271875	slug/ft
Beam elastic modulus	E	0.1584×10^{10}	lb/ft^2

Table 2: Natural frequencies

w_1	0	rad/s
w_2	6.454	rad/s
w_3	52.41	rad/s
w_4	1.607×10^2	rad/s
w_5	3.381×10^2	rad/s
w_6	5.78×10^2	rad/s

Flexible Satellite: Computational Efficiency

Coarser macro grids (higher p) and the **LF** formulation both shrink the optimisation problem by reducing the number of variables and constraints.

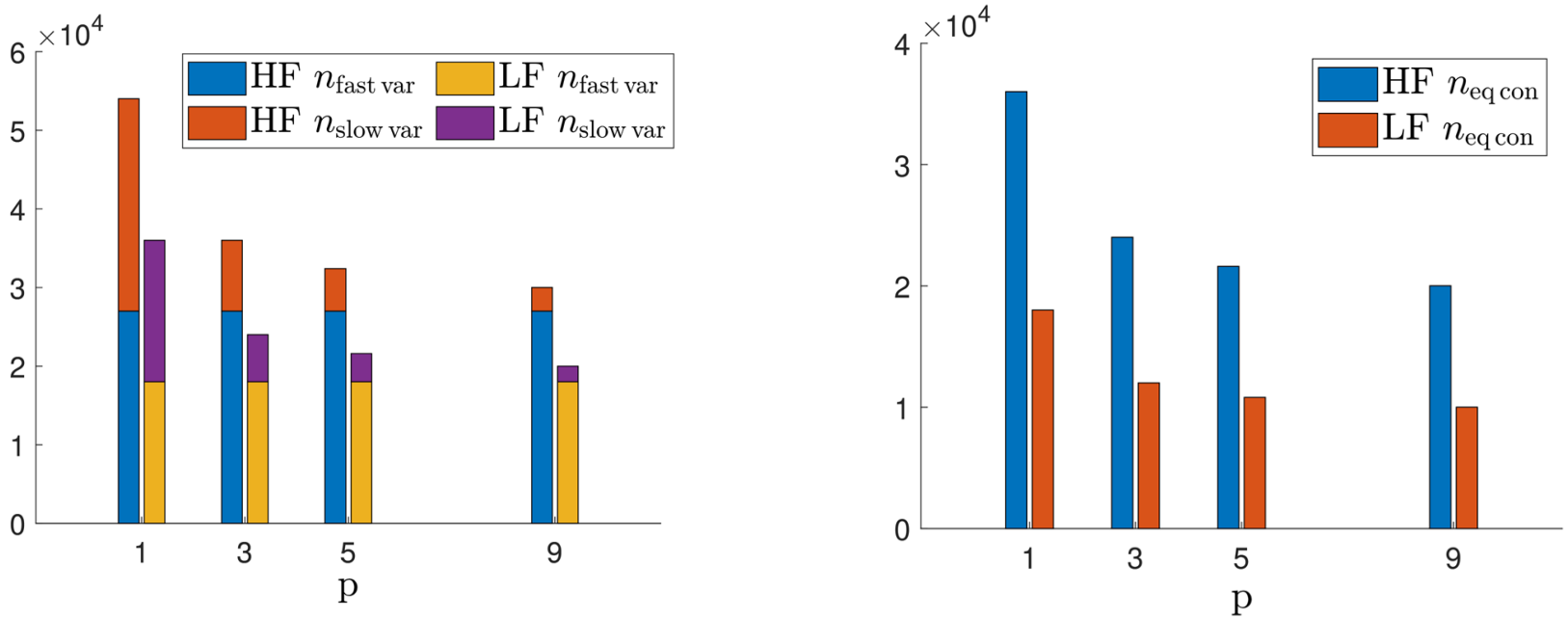
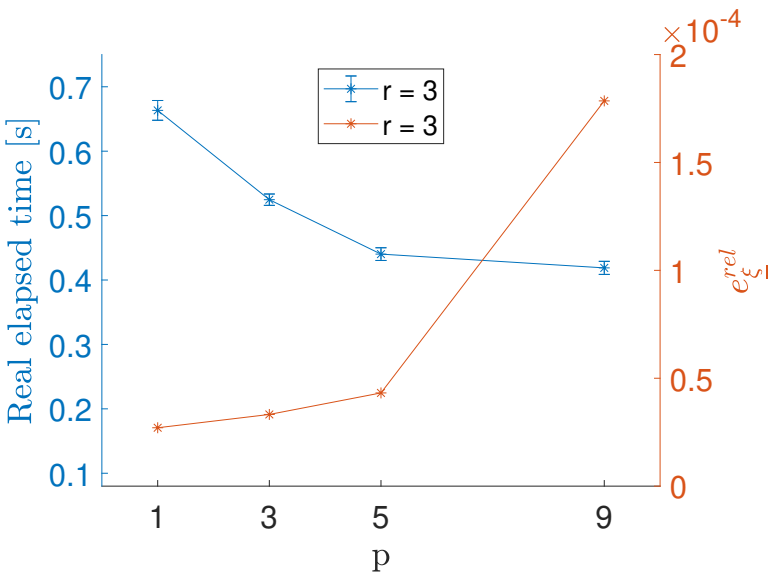


Figure 4: Size of the OCP using Multirate DMOC HF and LF with varying p, $t_f = 4.5$, $\Delta t = 10^{-3}$

Accuracy vs. Efficiency: HF vs. LF

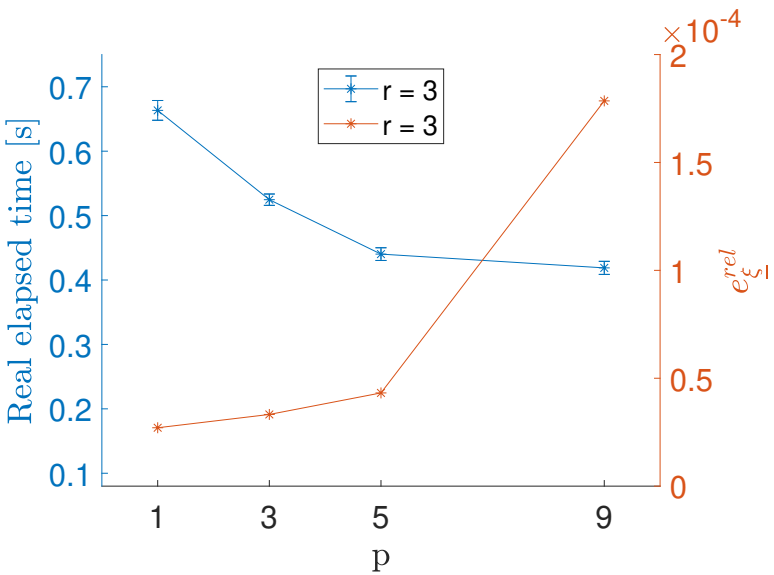
HF: real-elapsed time drops **34%** as p goes from 1 to 5. **LF** cuts runtime a further **61–63%**, at comparable accuracy.



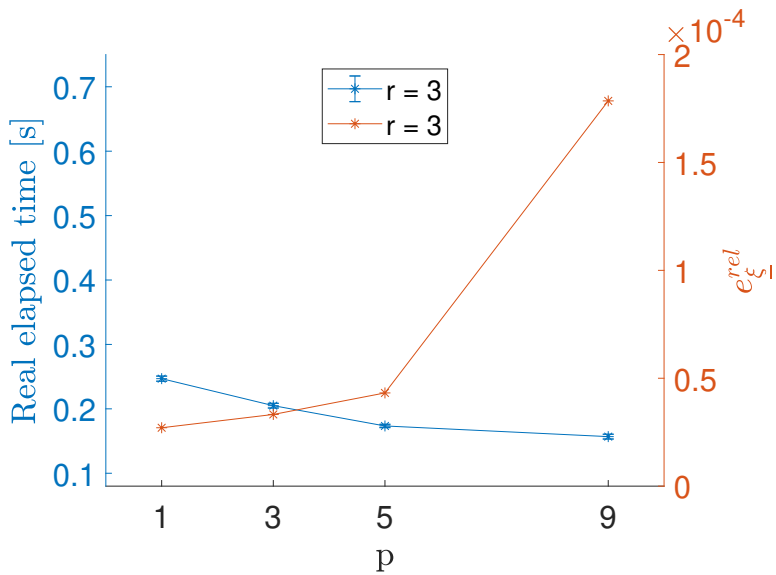
HF: runtime & relative error vs. p

Accuracy vs. Efficiency: HF vs. LF

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HF: runtime & relative error vs. p

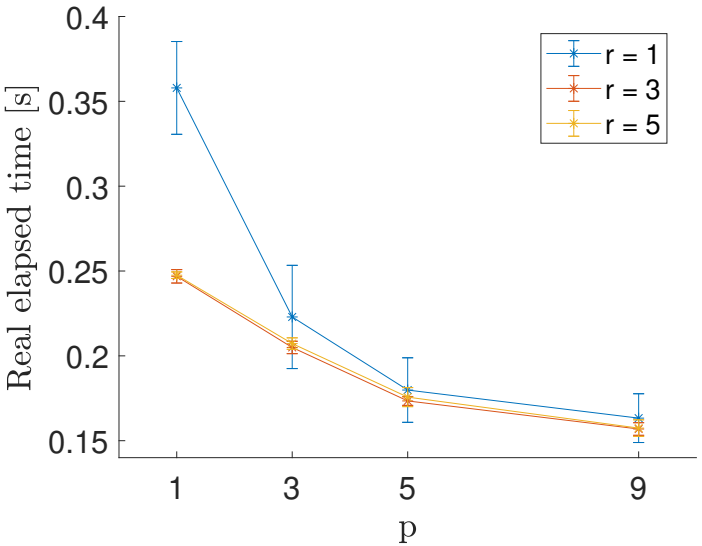


LF: runtime & relative error vs. p

Accuracy vs. Efficiency: Robustness to r

Runtime and accuracy trends are **robust to the choice of** r (slow-mode count)

$r = 1, 3, 5$ behave almost identically for $p \geq 3$.

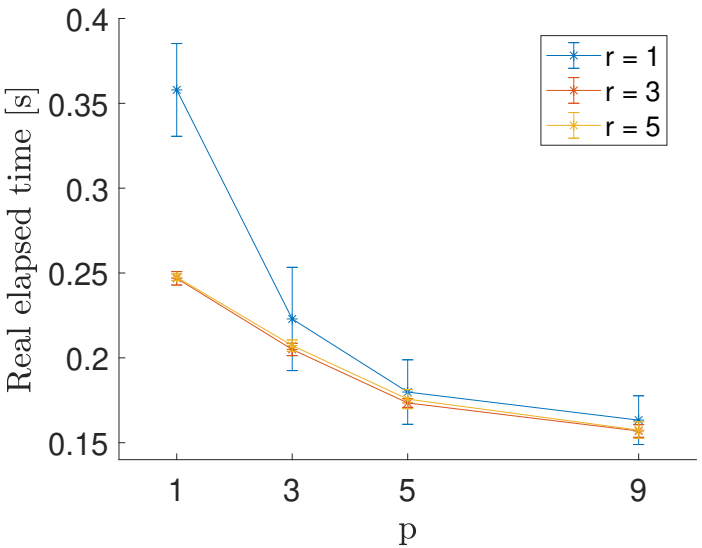


Mean runtime vs. p for $r = 1, 3, 5$

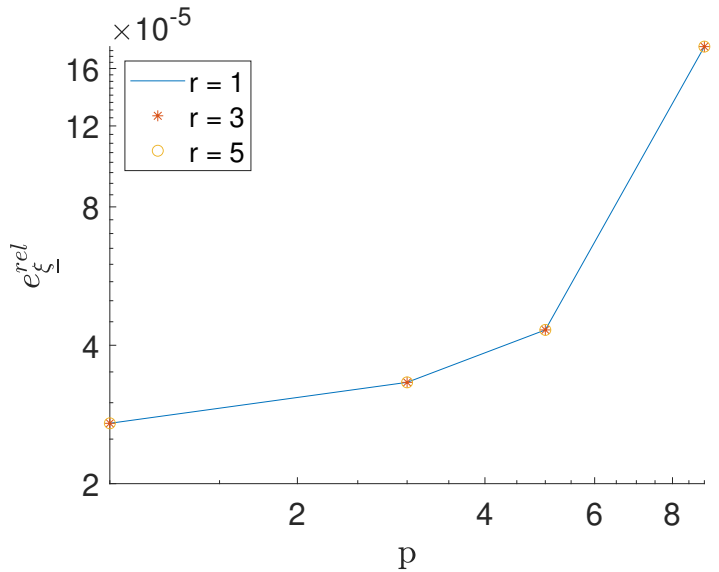
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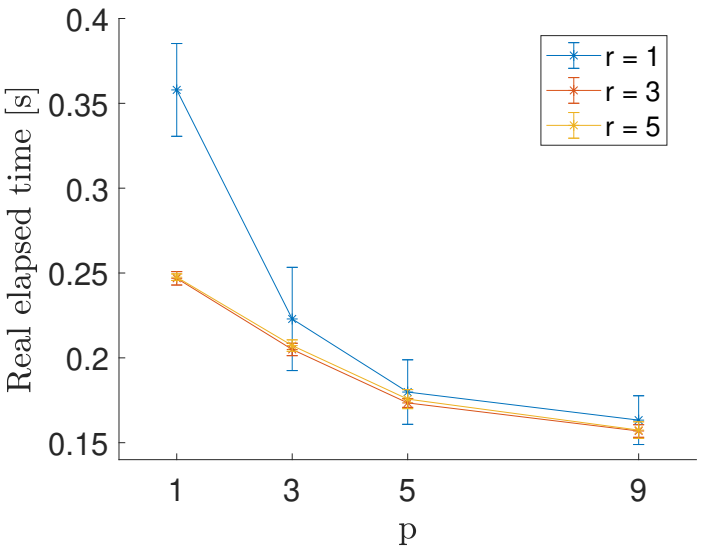


Relative error vs. p for $r = 1, 3, 5$

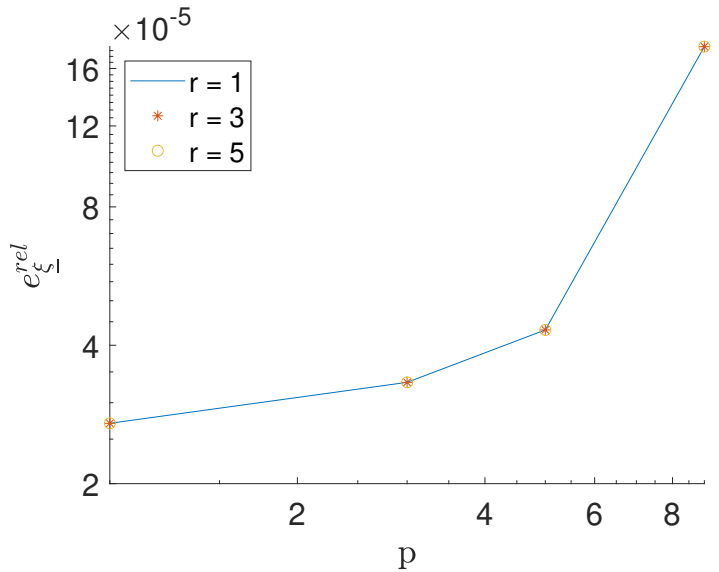
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Mean runtime vs. p for $r = 1, 3, 5$



Relative error vs. p for $r = 1, 3, 5$

In the paper we also numerically demonstrate the accuracy order and structure-preserving properties of this approch for flexible satellite control.

Conclusions

Multirate DMOC: High-fidelity simulation at a reduced computational cost

- ✓ Structure-preserving $\rightarrow$ *conservation of energy and/or momenta of the system*
- ✓ Multirate discretisation $\rightarrow$ *computational efficiency*
- ✓ Unified control methodology $\rightarrow$ *potential improvement in optimality and constraint-handling capabilities*
- ✓ State-space formulation $\rightarrow$ *further computational savings for equivalent accuracy, with structure preservation*
- ✓ No need to decouple the equations of motion $\rightarrow$ *simple application to a variety of linear and nonlinear models*
- ✓ Straightforward selection of slow/fast subsystems $\rightarrow$ *tailor the solution to the time scales of the dynamics for further computational savings*

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Continued work (see next slide for references):

- extension to a robust multirate variational receding horizon framework,
- extension to systems with kinematic constraints,
- theoretical justifications and quantification of the structure-preserving and accuracy properties of Multirate DMOC,
- extensions for improved inference of system dynamics,
- future work: effect of the truncated (unmodeled) vibration modes.

Thank You!

For questions please contact: Yana Lishkova, lishkova@caltech.edu

California Institute of Technology (Caltech), Pasadena, USA

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